RUIKE LYU

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Research Interests: Power and energy systems; demand flexibility from large loads; sustainable energy infrastructure planning; electricity markets.

Research Appointments

- **Department of Earth and Environmental Engineering, Columbia University** *Oct. 2026–Present*
  Postdoctoral Research Scientist, Postdoctoral Advisor: Prof. Bolun Xu New York, NY
- **Dept. of Electrical and Electronic Engineering, The Hong Kong Polytechnic University** *Jul. 2026–Sept. 2026*
  Postdoctoral Research Associate Hong Kong
- **Andlinger Center for Energy and the Environment, Princeton University** *Mar. 2025–Feb. 2026*
  Visiting Research Collaborator, Faculty Host: Prof. Jesse Jenkins Princeton, NJ

Education

- **Department of Electrical Engineering, Tsinghua University** *Sept. 2021–Jun. 2026*
  Ph.D. in Electrical Engineering, with the university's highest honor for graduates; PhD Advisor: Prof. Chongqing Kang Beijing, China
- **Department of Electrical Engineering, Tsinghua University** *Sept. 2017–Jun. 2021*
  B.E. in Electrical Engineering, with honors Beijing, China
- **School of Economics and Management, Tsinghua University** *Sept. 2018–Jun. 2021*
  Bachelor of Business Administration (Second Degree) Beijing, China

Selected Honors and Awards

- **Outstanding Doctoral Dissertation**, Tsinghua University *2026*
- **Finalist, Top 10 Rising Stars in Research**, Tsinghua University *2026*
- **Best Presentation Award (one of two)**, IEEE PES Ph.D. Dissertation Challenge *2025*
- **Best Oral Presentation Award**, PCCC Student Networking Session, IEEE PES General Meeting *2025*
- **National Scholarship for Graduate Students**, Ministry of Education of China *2025, 2023*
- **Finalist, University-wide Top 10 Students**, Tsinghua University *2025*
- **Outstanding Senior Thesis**, Tsinghua University *2021*

Publications

J=Journal Article, C=Conference Paper

Manuscripts under Review

[J.16] **R. Lyu** and H. Guo, "Demand flexibility must account for consumer-side capital costs," *Nature Reviews Clean Technology*, under review.

[J.15] **R. Lyu** and T. Li, "Do We Still Need Demand Flexibility as Batteries Become Cheaper? A Levelized Cost Perspective," *Energy Conversion and Management*, under review. arXiv:2608.07348.

Journal Articles

[J.14] **R. Lyu**, A. Li, J. Wang, H. Luo, E. Du, H. Guo, C. Kang, and J. Jenkins, "Industrial overcapacity can enable seasonal flexibility in electricity use," *Nature Energy*, **11**, 1203–1215 (2026). (**Cover Article**)

[J.13] Y. Shen, **R. Lyu**, S. Wang, J. Dai, H. Guo, and C. Kang, "Thermal-Mass Coupled Modeling and Optimal Scheduling of Aluminum Smelting Loads for Safe Demand Response Participation," *IEEE Transactions on Smart Grid*, in press.

[J.12] X. Su, **R. Lyu**, Y. Bai, X. Wang, W. Zhao, and H. Guo, "A Method for Modeling the Feasible Region of Industrial Production Processes Based on the Optimal Adjustable Load Model," *Automation of Electric Power Systems*, 2026, 50 (02): 18-29.

[J.11] **R. Lyu**, H. Guo, G. Strbac and C. Kang, "Data-Driven Dimension Reduction for Industrial Load Modeling Using Inverse Optimization," *IEEE Transactions on Smart Grid*, vol. 16, no. 3 (2025): 2695-2698.

[J.10] **R. Lyu**, H. Guo, Q. Tang, Q. Chen, and C. Kang, "Production Scheduling Identification: An Inverse Optimization Approach for Industrial Load Modeling Using Smart Meter Data," *IEEE Transactions on Smart Grid*, vol. 16, no. 2 (2025): 1207-1220.

[J.9] **R. Lyu**, X. Su, E. Du, H. Guo, Q. Chen and C. Kang, "Efficient Scheduling of Discrete Industrial Processes through Continuous Modeling," *IEEE Transactions on Smart Grid*, vol. 16, no. 6 (2025): 4726-4740.

[J.8] Q. Chen, **R. Lyu**, H. Guo, and X. Su, "Real-Time Operation Strategy of Virtual Power Plants With Optimal Power Disaggregation Among Heterogeneous Resources," *Applied Energy*, vol. 361 (2024): 122876.

[J.7] X. Su, **R. Lyu**, H. Guo, and Q. Chen, "A Method for Optimal Selection of High-Capacity Industrial Users for Demand Response Based on Load Step Data Processing Mode," *Electric Power*, 2024, 57(01): 18-29.

[J.6] **R. Lyu**, H. Guo, K. Zheng, M. Sun, and Q. Chen, "Co-Optimizing Bidding and Power Allocation of an EV Aggregator Providing Real-Time Frequency Regulation Service," *IEEE Transactions on Smart Grid*, vol. 14, no. 6 (2023): 4594-4606.

[J.5] Q. Chen, **R. Lyu**, H. Guo, H. Jia, Y. Ding, Y. Wang, and C. Kang, "Electricity User Behavior Modeling for Demand Response: Research Status Quo and Applications," *Electric Power Automation Equipment*, 2023, 43 (10): 23-37.

[J.4] H. Zhou, Q. Shao, X. Zhu, S. Liu, X. Liu, and **R. Lyu**[*], "An Incentive-Compatible Frequency Regulation Market for Flexible Resources in Microgrid," *IEEE Access*, vol. 11 (2023): 18983–18994.

[J.3] **R. Lyu**, Y. Gu, and Q. Chen, "Electric Vehicle Charging Right Trading: Concept, Mechanism, and Methodology," *IEEE Transactions on Smart Grid*, vol. 13, no. 4 (2022): 3094-3105.

[J.2] Q. Chen, **R. Lyu**, Q. Tang, K. Li, H. Gao, and H. Guo, "Emergency Response of Electricity Market from Perspectives of Suspension of Spot Market in Australia," *Automation of Electric Power Systems*, 2022, 46 (16): 214-223.

[J.1] Q. Chen, X. Fang, H. Guo, K. Zheng, Q. Tang, **R. Lyu**, K. Pan, P. Palensky, D. S. Kirschen, and C. Kang, "The Competition and Equilibrium in Power Markets Under Decarbonization And Decentralization," *iEnergy*, vol. 1, no. 2 (2022): 188-203.

Conference Papers

[C.10] J. Li, **R. Lyu**, Y. Zhang, X. Cha, K. Zheng, and H. Guo, "Value-Based Industrial Load Shedding with Supply Chain Coordination Reduces Electricity Cost," *2026 IEEE PES International Meeting*, Hong Kong, 2026, pp. 1-5.

[C.9] J. Zhang, **R. Lyu**, X. You, J. Wang, Y. Cai, and H. Guo, "When Will Real-Time Pricing Outperform Time-of-Use Pricing Significantly?," *2026 IEEE PES International Meeting*, Hong Kong, 2026, pp. 1-5.

[C.8] H. Huang, **R. Lyu**, C. Feng, H. Zhong, H. B. Gooi, B. Li and R. Liang, "Learning for Feasible Region on Coal Mine Virtual Power Plants with Imperfect Information," *2025 IEEE PES General Meeting*, Austin, TX, USA, 2025, pp. 1-5.

[C.7] Q. Liu, **R. Lyu**, Z. Zhai, Y. Shen, X. Liu and H. Guo, "Integrating Fast-response Capability into Virtual Power Plant Operation for Ancillary Services," *2025 IEEE Powertech*, Kiel, Germany.

[C.6] R. Chen, Z. Tang, **R. Lyu**, Q. Zheng, H. Song and H. Guo, "Combining AI and Simulation to Assess Building Demand Response Potential at Scale," *2025 5th International Conference on Advances in Electrical, Electronics and Computing Technology (EECT)*, Guangzhou, China, 2025, pp. 1-6. (**Best Presentation**)

[C.5] Y. Shen, **R. Lyu**, H. Guo and C. Kang, "An Improved Modeling Method for Electrolyte Aluminum Loads Considering Thermal Balance and Flexible Regulation Cost," *2025 IEEE International Conference on Power Systems and Smart Grid Technologies (PSSGT)*, Chongqing, China, 2025, pp. 331-337. (**Best Presentation**)

[C.4] **R. Lyu**, H. Guo and Q. Chen, "Approximating Energy-Regulation Feasible Regions of Virtual Power Plants: A Data-Driven Inverse Optimization Approach," *2024 IEEE Power & Energy Society General Meeting (PESGM)*, Seattle, WA, USA, 2024, pp. 1-5.

[C.3] A. Luo, **R. Lyu**, H. Guo, Y. Cai and Q. Chen, "An Incentive-Compatible VPP Profit Allocation Model Considering the Operating Characteristics of Air Conditioning Loads," *2024 7th International Conference on Energy, Electrical and Power Engineering (CEEPE)*, Yangzhou, China, 2024, pp. 1270-1278. (**Best Presentation**)

[C.2] L. Su, **R. Lyu**, H. Guo, Y. Cai and Q. Chen, "Coordinating Air Conditioning Load Clusters to Provide Frequency Regulation Using Setpoint Changes," *2024 7th International Conference on Energy, Electrical and Power Engineering (CEEPE)*, Yangzhou, China, 2024, pp. 905-910.

[C.1] **R. Lyu**, H. Guo, Y. Zheng, Y. Bai and Q. Chen, "LSTN: A Linear Model of Industrial Production Process for Demand Response," *2023 IEEE PES Innovative Smart Grid Technologies Europe (ISGT EUROPE)*, Grenoble, France, 2023, pp. 1-5.

SELECTED RESEARCH PROJECTS

- **Net Zero China** *2024–Present*
  Core Team Member; joint initiative of Tsinghua University and Princeton University
  - Develop sector- and region-specific industrial energy-demand and decarbonization scenarios using EnergyPlus.
  - Implement aluminum, ammonia, and methanol production models in the open-source MacroEnergy.jl energy-system planning framework.
- **Energy Conservation and Supply-Demand Interaction Technology for High-Energy-Consumption Industrial Users**
  Core Team Member; National Key Research and Development Program of China, Grant No. 2023YFB2407300, 2023–2026
- **Key Technologies of Interactive Regulation of Large-Scale Flexible Resources Aggregation in Virtual Power Plants**
  Core Team Member; National Key Research and Development Program of China, Grant No. 2021YFB2401200, 2021–2025

Industry Collaboration

- **Industrial Energy Management for a BYD Automotive Manufacturing Plant** *2024*
  Implementation Lead; State Power Investment Corporation (SPIC) Chongqing, China
  - Developed and implemented an optimization strategy coordinating a 60 MW manufacturing load, a 60 MW/240 MWh battery energy storage system, and 10 MW of solar generation.
  - Projected annual operating-cost savings of approximately US$4 million.

Patent Applications

[PAT.2] Q. Chen, Y. Chen, K. Zheng, and **R. Lyu**, "Method for updating state of charge based on power characteristic of electrochemical model of lithium-ion battery," U.S. Patent Application No. 18/919,638 (2025).

[PAT.1] Q. Chen, Y. Chen, K. Zheng, and **R. Lyu**, "Method for estimating state of power based on electrochemical model of lithium-ion battery," U.S. Patent Application No. 18/919,593 (2025).

Invited Seminars and Talks

- **The Chinese University of Hong Kong** *Sept. 17, 2026*
  Host: Prof. Yue Chen Hong Kong
- **The University of Hong Kong** *Apr. 1, 2026*
  Host: Prof. Yi Wang Hong Kong
- **Columbia University** *Feb. 9, 2026*
  Host: Prof. Bolun Xu New York, NY
- **Princeton University** *Feb. 6, 2026*
  Host: Prof. Denise Mauzerall Princeton, NJ
- **Harvard University** *Feb. 4, 2026*
  Host: Dr. Chris Nielsen Cambridge, MA
- **Massachusetts Institute of Technology (MIT)** *Feb. 3, 2026*
  Host: Dr. Ruaridh MacDonald Cambridge, MA
- **Dartmouth College** *Feb. 2, 2026*
  Host: Prof. Cong Chen Hanover, NH
- **New York University** *Jan. 28, 2026*
  Host: Prof. Yuzhang Lin New York, NY
- **Stanford University** *Dec. 5, 2025*
  Host: Prof. Adam Brandt Stanford, CA
- **University of Washington** *Oct. 17, 2025*
  Host: Prof. Baosen Zhang Seattle, WA
- **California Institute of Technology (Caltech)** *Oct. 16, 2025*
  Host: Prof. Steven Low Pasadena, CA
- **University of California, Riverside** *Sept. 26, 2025*
  Host: Prof. Nanpeng Yu Riverside, CA

Teaching and Mentoring

- **Department of Electrical Engineering, Tsinghua University** *2023–2024*
  Teaching Assistant for Energy Systems Operation, Dispatch, and Planning Beijing, China
  - Prepared course materials on energy storage, virtual power plants, and electricity markets; designed and graded all assignments.
  - Developed two open-ended optimization projects based on virtual-power-plant research, in which students formulated models and proposed their own solution methods.
- **Department of Electrical Engineering, Tsinghua University**
  Graduate Research Mentoring Beijing, China
  - Co-mentored three master's students and one PhD student.

Academic and Professional Service

- **Peer review:** Completed 40 reviews for 10 journals, including *IEEE Transactions on Smart Grid*, *Applied Energy*, *IEEE Transactions on Power Systems*, *IEEE Internet of Things Journal*, and *Scientific Reports*.
- **Professional service:** Member of the IEEE Task Force on Systems Modelling for Electricity Market Design and the IEEE Task Force on Economics of Flexibility Models & Data; co-hosted a task-force meeting at the 2026 IEEE Power & Energy Society General Meeting and a 2026 symposium in Montreal on energy and power education, research innovation, and career development in the AI era.
- **University service:** Served as an Undergraduate Counselor at Tsinghua University; conducted more than 80 one-on-one advising sessions, organized seven faculty discussions on research and careers, and coordinated industrial site visits. Recognized as an Outstanding Undergraduate Counselor.
- **Academic community engagement:** Served as a panelist at IEEE PES Day events and led presentations and student discussions on emerging energy policies and technologies through the Princeton University Energy Association.